

OpenClaw RGB Light Control Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-OpenClaw Prompts], [05-AI Voice Interaction Configuration], and [06-Three Conversation Scheme Configuration and Testing] in the `Preparation for Use` directory. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This document assumes the basic environment is already prepared and only expands on RGB-related content.

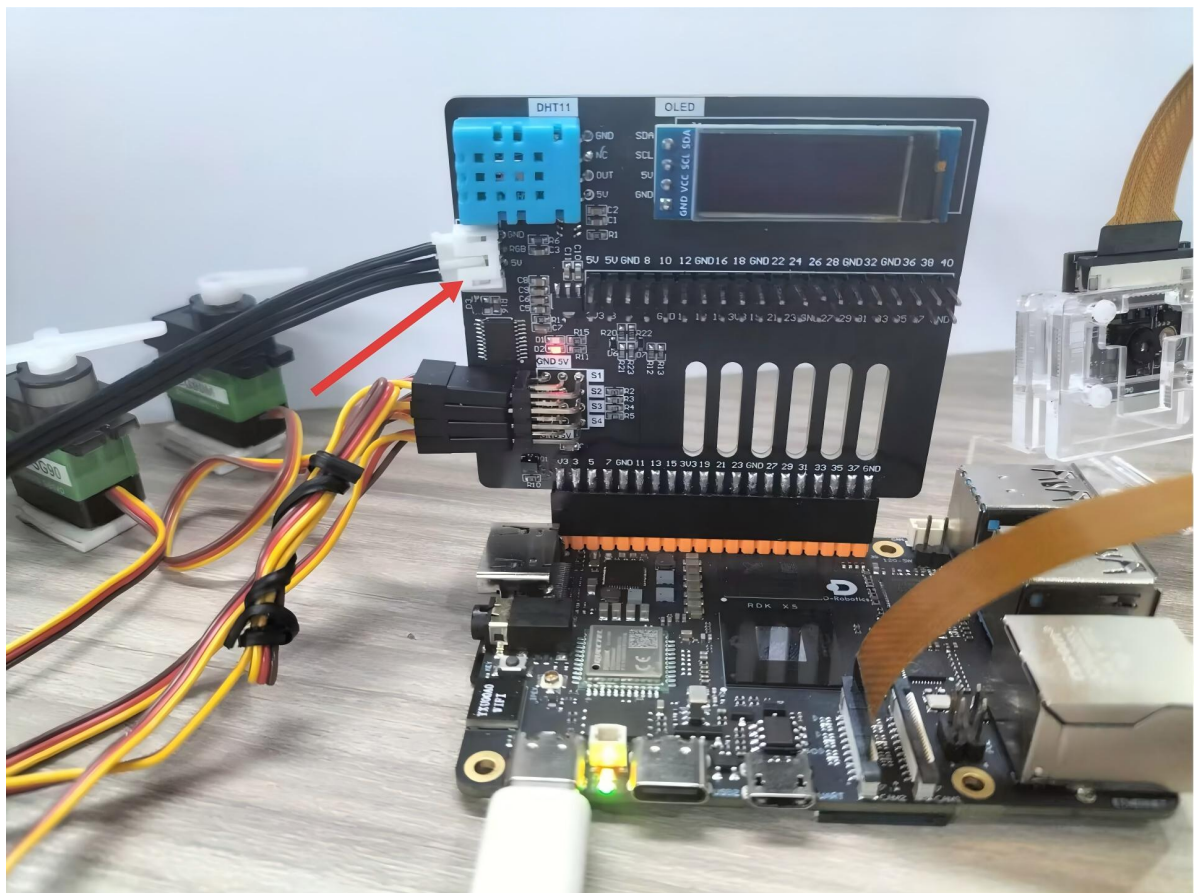
1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capabilities through RGB light peripherals. After completing this tutorial, readers will be able to:

- Understand the basic idea of OpenClaw controlling RGB lights
- Interact with OpenClaw via three methods: WhatsApp, TUI, and voice code
- Complete four RGB cases: single color lighting, breathing light, warning flash, rainbow marquee
- Understand the complete pipeline from "user input" to "light execution"

2. Hardware Preparation

- OpenClaw main board
- RGB light strip / RGB LED module [Default 14 WS2812 LEDs]
- [Optional] Microphone module
- Wiring diagram



You can add RGB wiring diagrams and hardware photos here.

3. Software Preparation

Code entry points used in the tutorial:

- Low-level driver code: `/home/sunrise/hardware_hal_demo/`
- RGB control command: `/home/sunrise/hardware_hal_demo/rgb`
- Voice entry point: `/home/sunrise/voice_to_openclaw/src/main.py`

The `rgb` tool already supports the following capabilities:

- Set fixed color `color`
- Set light effect `effect`
- Turn off lights `off`

Current common light effects include:

- `breathe` Breathing light
- `blink` Flashing
- `flow` Flowing light
- `marquee` Marquee light
- `rainbow` Rainbow light

If you want to verify the underlying RGB commands separately, you can also enter the following content in the terminal:

```
rgb color --color red # All turn red
rgb effect --effect breathe --color blue --speed 5 # Breathing light
rgb off # Turn off lights
```

For more commands, you can check the md documents in this folder:

```
/home/sunrise/hardware_hal_demo/README.md
```

If the low-level commands can be executed normally, it means the hardware connection and communication with the RDK X5 motherboard are normal.

4. Testing RGB Lights

Before starting this tutorial, please complete a basic `openclaw tui` conversation self-check first; if you haven't done so, see [06-Three Conversation Scheme Configuration and Testing]. After confirming that OpenClaw can respond normally, proceed to the RGB-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

```
OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws--type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

█
```

You can try entering the following content in the terminal:

- Set all RGB lights to red
- Change RGB lights to blue
- Lower the brightness of RGB lights a bit
- Turn off RGB lights

```
Set all RGB lights to red

All RGB lights set to red. 🟠🟠🟠

Change RGB lights to blue

All RGB lights changed to blue. 🟡🟡🟡

Turn off RGB lights

RGB lights turned off. •
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
44k/1.0m (4%)

█
```

If you want to test a specific LED, you can also try entering content like the following:

- Set the 1st LED to green
- Set the 5th LED to purple

5. Three Schemes for Conversing with OpenClaw

Scheme	Suitable Scenarios	Advantages	Suggested Positioning
WhatsApp	Remote control, classroom demonstration	Easy access, suitable for multi-user experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	Get the control pipeline running first
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration scheme

If you have selected the AI voice module + speaker, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to enable broadcast parameter

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal.
```

- Enter in terminal:

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Scheme

For WhatsApp access steps, please refer directly to [06-Three Conversation Scheme Configuration and Testing]. This section only retains conversation examples suitable for the RGB tutorial.

After creation, you can converse directly, for example:

- Change the light strip to blue
- Turn on rainbow light effect
- Make the lights flash faster

19:44



< 2



+86 153 3885 7526 (你)

给自己发消息

The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.

19:42 ✓✓

Done! The servo completed a left-to-right swing:

- Moved to 0° (left)
- Swung to 180° (right)
- Returned to 90° (center)

The servo is now ready for the next command.

19:42 ✓✓

Make the light strip blue 19:44 ✓✓

Done! The light strip is now blue at 50% brightness. All 14 LEDs are lit with a nice blue glow.

19:44 ✓✓



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M



123



space

return



5.2 TUI Scheme

If you have completed the TUI self-check in Part 6, you can continue the conversation by entering the following command in the terminal:

```
openclaw tui
```

```
OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
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18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)
```

5.3 Voice Code Scheme

Requires selection of AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [04-AI Voice Interaction Configuration].

```
cd /home/sunrise/voice_to_openclaw
source .venv/bin/activate
python -m src.main
```

After wake-up, you can directly speak control content, for example:

- Change the lights to red
- Turn on blue breathing light
- Make the lights flow like a rainbow


```
A 2026-05-14 14:56:02 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
C [系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
A [唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
T [听取] 正在收音
[识别] 已捕获 3780 ms 语音, 正在转写

[你] Take the lights red.
R [发送] 正在交给 OpenClaw
c [完成] OpenClaw 返回 200
a
4 [OpenClaw] Done! The RGB light strip is now red. 🟡

All 14 LEDs are glowing red
at 50% brightness.
```

6. Comprehensive Cases

Note: The following are conversation demonstrations conducted via `openclaw tui`. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Single Color Lighting

Experiment Objective

Let the RGB lights stably display a certain fixed color to complete the most basic light control.

Effect Description

After receiving the instruction, all LEDs of the RGB lights uniformly display the specified color, such as red, green, blue, white, or warm white. This case is suitable for verifying if color mapping is normal.

Example Instructions

- Set all RGB lights to red
- Change the light strip to blue
- Change the lights to warm white

6.2 Case 2: Breathing Light

Experiment Objective

Let the RGB lights perform brightness gradient on a certain color, forming a breathing light effect.

Effect Description

The RGB lights will cycle from dark to bright and then from bright to dark. The overall effect is relatively soft, suitable for standby prompts, atmosphere lights, or status indicator lights.

Example Instructions

- Turn on blue breathing light
- Change the lights to green breathing effect
- Slow down the breathing light speed a bit

6.3 Case 3: Warning Flash

Experiment Objective

Let the RGB lights simulate warning lights or alarm prompt effects through rapid flashing.

Effect Description

The lights can flash rapidly using red, yellow, or alternating red and blue, suitable for expressing information such as "attention," "warning," or "abnormal status," making the visual feedback more obvious.

Example Instructions

- Make the lights flash red
- Turn on yellow warning light
- Make the RGB lights flash quickly three times

6.4 Case 4: Rainbow Marquee

Experiment Objective

Let the RGB lights form a continuous flowing dynamic effect to showcase richer light performance.

Effect Description

The LED colors will change sequentially and flow forward, looking like rainbow flowing lights or marquee lights. This effect is very suitable for boot animations, display animations, or interactive light effects.

Example Instructions

- Turn on rainbow light effect
- Make the lights flow like a marquee
- Turn on blue flowing light, faster speed